

# Homework 3

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## 1 Two-level system

Let  $\varepsilon_0 < \varepsilon_1 \in \mathbb{R}$  and  $\gamma, \chi \in \mathbb{R}_{\geq 0}$ . Define the Hamiltonian

$$H = \begin{pmatrix} \varepsilon_0 & \gamma e^{-i\chi} \\ \gamma e^{i\chi} & \varepsilon_1 \end{pmatrix}.$$

Energy eigenvalues

$$(\varepsilon_0 - \lambda)(\varepsilon_1 - \lambda) - \gamma^2 = 0 \implies \lambda^2 - (\varepsilon_0 + \varepsilon_1)\lambda + \varepsilon_0\varepsilon_1 - \gamma^2 = 0$$

$$\lambda = \frac{\varepsilon_0 + \varepsilon_1 \pm \sqrt{\varepsilon_0^2 + \varepsilon_1^2 + 2\varepsilon_0\varepsilon_1 - 4\varepsilon_0\varepsilon_1 + 4\gamma^2}}{2} = \frac{\varepsilon_0 + \varepsilon_1 \pm \sqrt{(\varepsilon_0 - \varepsilon_1)^2 + 4\gamma^2}}{2}???.$$